UNIVERSITY OF DERBY

5CM506
Data Driven Systems

Normalisation

Dr. Maqbool Hussain

Aims of the Session

This session aims to help you:

1. Apply the theoretical concepts to produce a normalised database

Overview

- **More on Normalisation**
- Exercise
- Q&A

More on 3NF

A transitive dependency is of the form:

A → B
B → C
Therefore, A → C

Title → Genre
Title → Author
Author → AuthorNationality

Therefore, Title → AuthorNationality

Title	Genre	Author	AuthorNationality
Twenty Thousand Leagues Under the Sea	Sci Fi	Jules Verne	French
Journey to the Centre of the Earth	Sci Fi	Jules Verne	French
Leaves of Grass	Poetry	Walt Whitman	USA
Anna Karenina	Fiction	Leo Tolstoy	Russian
A Confession	Autobiography	Leo Tolstoy	Russian

More on 3NF

Given:

Title → Genre

Title → Author

Author → AuthorNationality

Therefore,

Title → AuthorNationality

The transitive dependency…

Title → Author → AuthorNationality

…needs to be removed.

<u>Title</u>	Genre	Author	AuthorNationality
Twenty Thousand Leagues Under the Sea	Sci Fi	Jules Verne	French
Journey to the Centre of the Earth	Sci Fi	Jules Verne	French
Leaves of Grass	Poetry	Walt Whitman	USA
Anna Karenina	Fiction	Leo Tolstoy	Russian
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More on 3NF

Given:

Title → Genre

Title → Author

Author → AuthorNationality

Therefore,

Title → AuthorNationality

The transitive dependency…

Title → Author → AuthorNationality

…needs to be removed.

Books (<u>Title</u>, Genre, Author fk)
Authors (<u>Author</u>, AuthorNationality)

<u>Title</u>	Genre	Author	AuthorNationality
Twenty Thousand Leagues Under the Sea	Sci Fi	Jules Verne	French
Journey to the Centre of the Earth	Sci Fi	Jules Verne	French
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A Confession	Autobiography	Leo Tolstoy	Russian

Overview

- More on Normalisation
- **Exercise**
- Q&A

Normalisation: Exercise

Given the following table:

1. Identify the functional dependencies
2. Identify the primary key
3. Convert the table to 2NF if necessary

SNO	SNAME	UNITNO	UNITNAME	LECTURER
0001	P Jones	PR1001	Programming	AJ Tyrell
0001	P Jones	SN1001	Computer Systems	P Marsden
0007	J Smith	PR1001	Programming	AJ Tyrell
0011	J Peters	PR2001	Data Structures	Z Atlass
0013	J Chan	PR2001	Data Structures	Z Atlass
0016	K Daley	PR1001	Programming	AJ Tyrell
0016	K Daley	PR2001	Data Structures	Z Atlass

Normalisation: Exercise

SNO -> SNAME
SNO -> UNITNO
SNO -> UNITNAME
SNO -> LECTURER
SNAME -> SNO
SNAME -> UNITNO
SNAME -> UNITNAME
SNAME -> LECTURER
UNITNO -> SNO
UNITNO -> SNAME
UNITNO -> UNITNAME
UNITNO -> LECTURER
UNITNAME -> SNO
UNITNAME -> SNAME
UNITNAME -> UNITNO
UNITNAME -> LECTURER
LECTURER -> SNO
LECTURER -> SNAME
LECTURER -> UNITNO
LECTURER -> UNITNAME

Given the following table:

1. **Identify the functional dependencies**
2. Identify the primary key
3. Convert the table to 2NF if necessary

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0016	K Daley	PR1001	Programming	AJ Tyrell
0016	K Daley	PR2001	Data Structures	Z Atlass

Normalisation: Exercise

SNO -> SNAME
UNITNO -> UNITNAME
UNITNO -> LECTURER

Given the following table:

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0016	K Daley	PR2001	Data Structures	Z Atlass

Normalisation: Exercise

SNO -> SNAME
UNITNO -> UNITNAME
UNITNO -> LECTURER

The primary key will consist of the left side of the functional dependencies:

SNO, UNITNO

Given the following table:

1. Identify the functional dependencies
2. **Identify the primary key**
3. Convert the table to 2NF if necessary

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Normalisation: Exercise

SNO -> SNAME
UNITNO -> UNITNAME
UNITNO -> LECTURER

We have a partial functional dependence, so we need to convert to 2NF as:

STUDENT_UNITS (SNO fk, UNITNO fk)
STUDENTS (SNO, SNAME)
UNITS (UNITNO, UNITNAME, LECTURER)

Given the following table:

1. Identify the functional dependencies
2. **Identify the primary key**
3. Convert the table to 2NF if necessary

SNO	SNAME	UNITNO	UNITNAME	LECTURER
0001	P Jones	PR1001	Programming	AJ Tyrell
0001	P Jones	SN1001	Computer Systems	P Marsden
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0016	K Daley	PR2001	Data Structures	Z Atlass

Overview

- More on Normalisation
- Exercise
- **Q&A**

Q&A

Things you might want to consider before submitting:

- Follow the **pre-session asynchronous tasks** on Study Materials -> Week 9-11
- Look at the **tutorial** to ensure your backend and frontend are linked: connecting SQL Server and MS Access
- Make sure you follow the requirements for submission (**ZIP folder** with the 3 components or you will fail the project)

Any questions?

Aims of the Session

You should now be able to:

1. Apply the theoretical concepts to produce a normalised database

References

Silberschatz, A., Korth, H. F., and Sudarshan, S. (2010), *Database System Concepts*, 6th Edition. McGraw-Hill.

- Chapter 8 – Relational Database Design (Sections 8.1, 8.2, 8.3 and 8.8)

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